

Higher Education and Household Financial Asset Allocation: Evidence from China’s University Enrollment Expansion

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Abstract

To explore the impact of higher education on household financial assets, this paper utilises data from the China Household Finance Survey (CHFS) and employs Fuzzy Regression Discontinuity Design (Fuzzy RDD) to leverage the quasi-experiment provided by China’s university enrollment expansion policy. The findings reveal that higher education significantly boosts household stock market participation, particularly among urban households, with negligible effects for rural households. Additionally, the influence of higher education on the stock market participation is more pronounced in families owning a single property compared to those with multiple properties. Mechanism analysis shows that higher education can influence the level of participation in household risk financial markets through two channels: improving individual financial literacy levels and income levels. These results underscore the critical role of higher education in facilitating stock market participation and contributing to wealth accumulation.

Keywords: higher education; stock market participation; university enrollment expansion; fuzzy regression discontinuity design

1 Introduction

Participating in financial markets and utilizing diverse investment strategies can increase personal wealth and hedge against risks, offering opportunities for capital appreciation and income growth. Besides, widespread public participation in financial markets can enhance asset allocation efficiency, stimulate the market’s price discovery function, and promote stable economic growth. However, the phenomenon of ‘limited participation’ is prevalent globally (Briggs et al., 2021). According to a report released by the Statista Research Department, 62% of U.S. households invested in the stock market in 2024. The participation rate remains below the pre-Great Recession peak, which reached 65% in 2007. Despite this, the proportion significantly exceeds that of China. According to data released by China Securities Depository and Clearing Corporation Limited, as of December 2024, the number of investors in China’s securities market reached 220 million accounts, representing only 15.6% of the national population. This relatively low participation rate reflects the broader trend of limited household engagement in financial markets. The 2019 edition of the China Household Finance Survey (CHFS) indicates that Chinese households hold a disproportionately large share of their wealth in real estate assets, which account for 66.3% of total household assets, while financial assets—including stocks, bonds, funds, and wealth

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management products—constitute only 12.8% (Xing et al., 2023). Moreover, within financial assets, Chinese households show a strong preference for low-risk instruments such as bank deposits and cash. Risky assets occupy only a small fraction of household portfolios: stocks, the most commonly held risky asset, represent just 15.45% of financial holdings, and the combined share of stocks, funds, and bank wealth management products remains below 25%. Participation in other riskier financial instruments is even more limited, with only 4.24% of households investing in mutual funds and less than 1% participating in the bond and derivatives markets (Gan et al., 2015).

Insufficient allocation of financial assets can trigger a series of issues. For households, a higher proportion of risk-free assets can diminish their ability to hedge against inflation risks. Variations in decision-making abilities within households significantly contribute to the substantial wealth inequality among households after retirement (Lusardi et al., 2017). When allocating assets, a reasonable level of risk can significantly enhance family happiness (Qiao and Cai, 2023). Therefore, lack of participation in financial markets can affect the improvement of households’ sense of well-being. On the other hand, allocating fewer family assets to risky financial assets can also impact the macro-economy (Deng et al., 2021). Firstly, it will increase financing constraints for enterprises, limit their ability to expand production, and hinder economic growth. Secondly, the low proportion of individual investors nationwide, will lead to insufficient information acquisition in the financial markets, which in turn increases financial market risk. The above analysis indicates that the allocation of risky financial assets in households affects not only their utility but also the healthy development of financial market. Therefore, understanding how to enhance household participation in risky assets is of great research significance.

Addressing this issue, this paper thoroughly investigates how the factor of education influences family asset allocation, with China serving as the focal point of the study. Education plays a pivotal role in influencing household investments in risky financial assets. Christiansen et al. (2008) believe that education enhances individuals’ ability to search, filter, and analyze information by exercising their financial cognitive level, thereby improving their decision-making in household risk financial asset allocation, boosting the participation rates in the stock market and stock returns; Calvet and Sodini (2014) find that education increases human capital accumulation, thereby increasing the likelihood of individuals choosing to allocate more financial risk assets. Wu and Shen (2015) suggest that higher-educated households tend to have higher incomes and greater risk-bearing capacity. Education, by elevating household income levels and improving social networks, enhances households’ resilience to risk and motivates them to invest in high-risk, high-return stock markets. Therefore, education level is closely related to household risk investments. China implemented the policy of University Enrollment Expansion in 1999, leading to a significant change in the number of people receiving higher education before and after the policy’s implementation. This created an well environment for conducting a quasi-natural experiment. Therefore, this paper selects China as the subject of its study.

Current research primarily utilizes the impact of the Compulsory Education Law as an instrumental variable for educational attainment to study the causal relationship between education and household allocation of risky financial assets. Gray et al. (2021) used two compulsory education reforms in the UK as treatment variables and employed a regression discontinuity design to explore the relationship between education and financial decisions. Ren et al. (2020) used the implementation of the Compulsory Education Law as an instrumental variable and found that educational attainment could increase the likelihood of families participating in the stock market. Black et al. (2018) and Cole et al. (2014), using data from the USA and Sweden respectively, discovered that the implementation of the Compulsory Education Law boosted the overall educational level of the population and that the increase in years of education prompted by the law could enhance financial market participation. However, using the implementation of the Compulsory Education Law as a quasi-natural experiment only verifies the impact of primary education development on household allocation of risky financial assets, while the influence of higher education

development remains to be studied. Compared to the development of primary education triggered by the Compulsory Education Law, the expansion of higher education due to university enrollment expansion policies is likely to have a more significant impact on the overall societal level of financial knowledge or income levels. Therefore, utilizing the university expansion policy can provide a better understanding of the causal relationship between the level of education and household allocation of risky assets.

The contribution of this paper lies in its novel use of China’s university expansion policy to estimate the causal relationship between higher education and the allocation of risky financial assets in Chinese households. And delved deeper into the impacts on two aspects: household financial market participation and household risky asset allocation. Building on this analysis, the paper further explores the channels through which higher education influences the allocation of risky financial assets in households. It posits that higher education can affect the allocation of risky assets through two main avenues: enhancing financial literacy and increasing income levels. Lastly, the paper conducts a heterogeneity analysis by differentiating between urban and rural areas and the number of housing properties owned, discussing whether the impact effects vary under the dual economic structure of China and different housing asset allocation backgrounds.

2 Theoretical Framework and Research Hypotheses

Household asset allocation refers to the process of distributing investment resources across different asset categories based on individual investment needs and objectives. In 1952, Markowitz introduced portfolio theory and developed the efficient frontier model, which posits that individuals make investment choices by balancing expected returns and associated risks. However, this theory rests on the assumption that investors have access to complete information and make fully rational decisions. In reality, many households allocate only a small portion—or none at all—of their assets to risky investments. Several factors have been identified to explain this limited participation, including income levels (Vissing-Jorgensen, 2002), risk preferences (Campbell, 2006), human capital (Bodie et al., 1992), and financial literacy (Van Rooij et al., 2011). Badarinsa et al. (2016) analyzed household financial behavior across 13 developed countries and found that investment experience, financial literacy, wealth levels, and interpersonal trust significantly influence household financial asset allocation. Similarly, Wang and Tian (2012) found that age, income, and education level are positively associated with stock market participation among urban residents in China. Notably, many of these influencing factors are closely linked to education. Higher educational attainment often leads to increased income, which in turn facilitates greater participation in financial markets. Moreover, education enhances financial literacy and shapes individuals’ risk attitudes, thereby indirectly affecting investment behavior. Therefore, the impact of education on household allocation of risky assets may operate through both direct and indirect channels.

This paper identifies financial literacy and income level as key mediating variables and conducts a mechanism analysis to explore these pathways in greater detail. Firstly, education can improve household risk asset allocation by improving residents’ financial literacy. In the study of the impact mechanism of education on household financial market participation and asset selection, Yin et al. (2014) find that higher education levels lead to greater financial knowledge within households, which drives households to participate more in risky stock markets and increase their allocation to risk assets. Vissing-Jorgensen (2002) argues that education is one of the main ways to influence financial cognitive levels: individuals with higher education can access more financial knowledge, have a greater ability to master financial knowledge, and are more likely to participate in the high-risk, high-return stock market. Frederick (2005) and Dohmen et al. (2010) use IQ data as a proxy for cognitive ability and find that individuals with higher IQs are more likely to participate in the stock market. Therefore, higher education levels are associated with higher financial cognitive levels, increasing the likelihood of household participation in

risk asset investment.

Secondly, higher education can enhance household allocation of risky financial assets by increasing income. [Cooper and Zhu \(2016\)](#) noted that education influences participation in financial markets by boosting household income. On one hand, individuals with higher educational qualifications tend to have higher earnings and job stability, which in turn enables them to take on the risks associated with the market for risky assets ([Wu and Shen, 2015](#)). On the other hand, financial literacy and financial interest are associated with higher risk tolerance ([Hermansson and Jonsson, 2021](#)). Individuals with higher education tend to have higher incomes, and those with higher incomes are more capable and motivated to diversify their wealth into high-risk, high-return assets like the stock market ([Ashenfelter and Rouse, 1998](#)). Therefore, this paper selects financial literacy and income level as the two variables to study the mechanisms affecting household allocation of risky assets.

Due to the fact that the allocation of household risk assets is a behavior that involves household decision-making and is closely related to individual's behavioral patterns. Therefore, the allocation of household risk assets can be measured not only by the investment ratio of risk assets, but also by whether the household participates in the financial market behavior ([Tsang and Yu, 2023](#)). Based on this understanding, this study explores the situation of household risk asset allocation from two dimensions: household financial market participation and household risk asset allocation. Furthermore, as risk financial assets mainly include stocks, funds, bonds, etc., and stocks are usually a more familiar and high proportion of risk asset categories for investors, this paper divides household risk asset allocation into two parts: overall risk asset allocation and specific stock asset allocation for detailed research. Through this analytical framework, this article aims to gain a deeper understanding of how households make risk asset allocation decisions based on different factors, and explore how these decisions affect the financial situation and investment behavior of households. The theoretical analysis framework and research hypotheses are as follows:

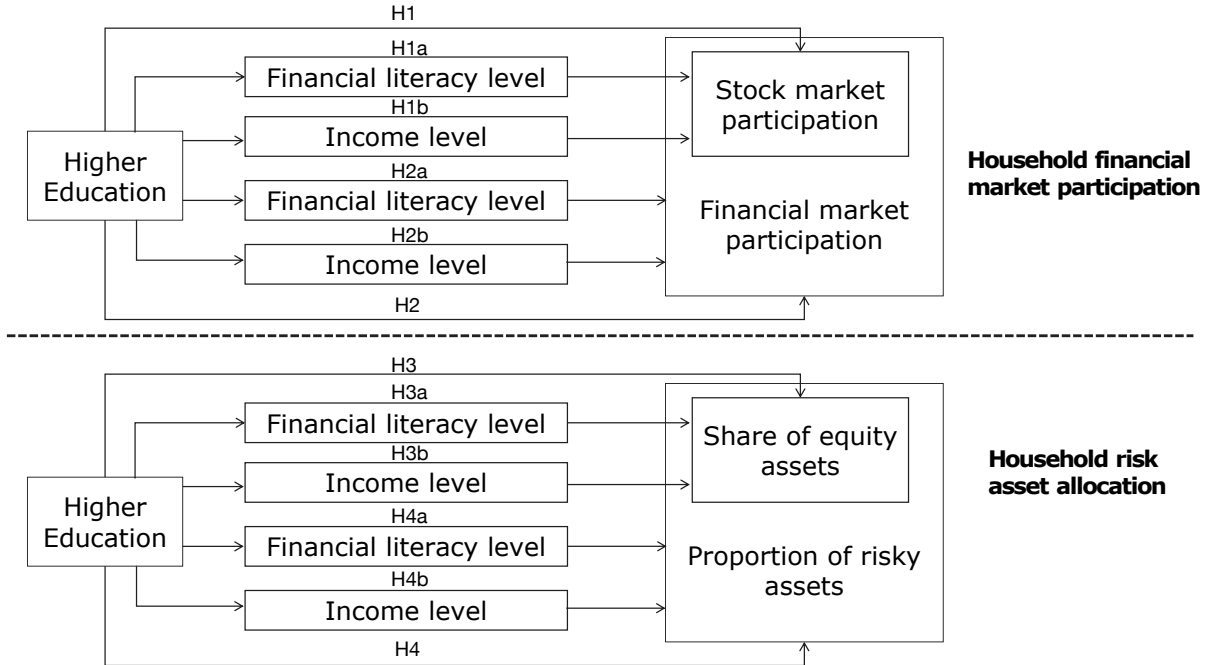


Figure 1: Theoretical Framework of Higher Education and Household Financial Asset Allocation

H1: Higher education enhances household participation in the stock market.

H1a: Higher education increases household stock market participation by improving financial literacy levels.

H1b: Higher education increases household stock market participation by increasing income levels.

H2: Higher education enhances household participation in financial markets.

H2a: Higher education increases household financial market participation by improving financial literacy levels.

H2b: Higher education increases household financial market participation by raising income levels.

H3: Higher education increases the proportion of equity assets in household portfolios.

H3a: Higher education enhances the household equity asset ratio by improving financial literacy levels.

H3b: Higher education increases the household equity asset ratio by raising income levels.

H4: Higher education increases the proportion of risky assets in household portfolios.

H4a: Higher education increases the household risky asset ratio by enhancing financial literacy.

H4b: Higher education increases the household risky asset ratio by increasing income levels.

3 Research Methods

3.1 Model selection

The choice of fuzzy regression discontinuity design (RDD) is predicated on the characteristics of the intervention at the threshold. Unlike sharp RDD, where the probability of receiving treatment jumps from 0 to 1 at the threshold, fuzzy RDD features a probability jump from a to b (where $0 < a < b < 1$) at the threshold. In the context of this paper, the university expansion policy does not guarantee university admission for all affected individuals, as admission also depends on scores, choice of major, and the region of the candidates, among other factors. Moreover, there exists a portion of the population that will not choose to attend university regardless of policy changes, while another portion will attend university no matter what. In this context, the higher education expansion policy has increased the probability of university enrollment for individuals who originally had no opportunity to attend, but this probability increase is not absolute (i.e., less than 1). When individuals cannot freely choose their treatment status, this discontinuous jump helps address endogeneity issues in better estimating the impact of higher education on household risky financial asset allocation.

Therefore, this paper employs a fuzzy RDD approach, using the 1999 higher education expansion in China as a quasi-natural experiment to estimate the impact of higher education on household risky financial asset allocation. After the policy implementation, a clear discontinuity in college enrollment rates between affected and unaffected groups was observed, leading to corresponding changes in their risky financial asset allocation. The fuzzy RDD approach estimates the impact of higher education by comparing changes in risky financial asset allocation between these two groups post-policy implementation.

3.2 Data source

The data used in this paper are sourced from the nationwide surveys conducted by the China Household Finance Survey and Research Center (CHFS) at Southwestern University of Finance and Economics, encompassing data from the years 2015, 2017, and 2019. CHFS provides detailed micro-level information on household finances, including demographic characteristics data that covers basic information of surveyed household members, such as age, gender, occupation, and income. CHFS also offers data on household assets and liabilities, including participation in the stock market and the allocation of risky financial assets like stocks, funds, and bonds.

This paper uses the date of birth as the sorting variable, identifying the cohort born in 1982 as the critical birth cohort (since the university expansion policy began in 1999). The sample size for individuals

born in 1982 or later is 6,881, while for those born before 1982, it is 15,159. In the data analysis, particular attention is paid to ensuring that under the optimal bandwidth condition, the selected samples are all born in 1971 or later. This approach ensures that all samples benefited from the educational level improvement brought by the Compulsory Education Law. Both the university expansion and the Compulsory Education Law, as universal education policies, could influence household allocation of risky financial assets. The Compulsory Education Law was enacted in 1986, affecting those born in 1971 or later. To exclude the possibility that changes in household risky financial asset allocation might stem from the Compulsory Education Law, this study focuses on samples born in 1971 or later under the optimal bandwidth condition, ensuring all samples are influenced by the policy, and the estimation results will not show significant differences due to the impact of the Compulsory Education Law. In subsequent empirical analyses, the samples selected under the optimal bandwidth condition are all born in 1971 or later. The optimal bandwidth meets the requirements for sample size and local regression in empirical analysis and aligns with the criterion that all samples are affected by the Compulsory Education Law.

Explained Variable: Household Allocation of Risky Financial Assets. Drawing on the methodologies of Zong et al. (2015) and Ren et al. (2020), this study assesses household allocation of risky financial assets from two dimensions: participation in the financial market and the configuration of risky assets.

Participation in the financial market is bifurcated into engagement with risky assets and stock market participation. Stock market participation is indicated by whether a household owns stocks; if a household possesses a stock account, then the variable `D_stock` is set to 1, otherwise, it is 0. Risky asset participation denotes whether a household holds any financial market risky assets; if so, the variable `D_all_assets` is assigned a value of 1, otherwise, it is 0.

The allocation of risky assets is quantified by the proportion of stock assets (`Equity_share`) and the proportion of all risky assets (`All_assets_share`) within the household's total financial assets. The stock asset ratio is the share of a household's stock assets in its total financial asset value, while the risky asset ratio represents the share of all risky financial investments in the household's total financial assets. Following the approaches of Yin et al. (2014) and Ren et al. (2020), financial assets are categorized into risk-free and risky assets. Risky financial assets in the market primarily include stocks, funds, bonds, financial derivatives, financial management products, foreign exchange, and gold; whereas risk-free financial assets comprise cash, cash in stock accounts, demand deposits, and time deposits. This classification and measurement approach aims to provide a nuanced analysis of the impact of higher education on household allocation of risky financial assets.

Treatment Variable: Higher education. Higher education is defined as a dummy variable: individuals with a final educational attainment of junior college or higher (college) are assigned a value of 1; those with educational attainment below junior college are assigned a value of 0.

Running variable: The birth year corresponding to the university expansion policy. The university expansion policy was implemented in 1999. Assuming all individuals in the sample started primary school at age 6 and did not repeat any grades, they would be 18 years old when taking the college entrance examination. This implies that the 1999 university expansion policy would affect individuals born in 1982 or later. Therefore, we use the date of birth as the sorting variable, with individuals born in 1982 or later forming the treatment group and those born before 1982 forming the control group. If an individual's birth year is post-1982, they are considered to be influenced by the university expansion and are placed in the experimental group; if born before 1982, they are deemed unaffected by the policy and are placed in the control group.

Control Variables: The control variables encompass two categories:

1. Demographic characteristics, which include age, gender (male or female), and marital status (married or unmarried);

2. Household characteristic variables, which encompass total household assets and annual household income ¹.

4 Empirical Analysis

4.1 Model

This paper adopts a two-stage least squares (2SLS) approach to estimate the causal effect within the framework of a fuzzy regression discontinuity design (FuzzRDD). In this context, the treatment variable—whether an individual has received higher education ($college_i$)—is not perfectly determined by the assignment variable. Instead, the implementation of the 1999 university enrollment expansion policy serves as an instrumental variable ($expansion_i$), indicating whether an individual was likely affected by the policy based on their birth cohort.

The first-stage regression equation is specified as:

$$college_i = \alpha_0 + \alpha_1 expansion_i + \alpha_2 X_i + \mu_i \quad (1)$$

The second-stage regression equation is given by:

$$y_i = \beta_0 + \beta_1 \widehat{college}_i + \beta_2 X_i + \varepsilon_i \quad (2)$$

Where: y_i denotes the outcome variable, capturing various dimensions of household risky financial asset allocation, including financial market participation, stock market participation, the proportion of risky assets, and the proportion of stock assets. $college_i$ is a binary indicator equal to 1 if the individual has received higher education, and 0 otherwise. $\widehat{college}_i$ is the predicted value from the first stage. $expansion_i$ is a binary instrumental variable equal to 1 if the individual was exposed to the university enrollment expansion policy, and 0 otherwise. X_i is a vector of control variables, including gender (male = 1), marital status (married = 1), rural residence (rural = 1), the logarithm of the household head's income, and the logarithm of total household assets. μ_i and ε_i are the error terms in the first and second stages, respectively.

This two-stage setup allows for the estimation of the local average treatment effect (LATE) of higher education on household financial behavior, addressing the endogeneity arising from the non-random assignment of education.

4.2 Descriptive Statistics

Table 1 provide an intuitive comparison between the control group and the treatment group in terms of educational attainment and financial market participation. The control group was not affected by the college enrollment expansion policy, while the treatment group was influenced by this policy. The specific analysis of the descriptive statistics is as follows:

Among all samples, the proportion of individuals with an junior of higher education level was 17.2%. In the control group, this proportion was 14.8%, while in the treatment group, it was significantly higher at 37.8%. This substantial difference suggests that the college enrollment expansion policy may have significantly improved educational attainment. Financial market participation was measured using two indicators: whether an individual holds a stock account and whether they participate in financial market investments. The descriptive statistics show that the treatment group significantly outperformed the control group on both indicators, which may indicate that individuals with higher education are more

¹This control variable will be removed in mechanism analysis.

inclined to participate in financial markets. Risk asset allocation was measured by two indicators: the proportion of stock assets to financial assets and the proportion of financial market risk assets to financial assets. The differences between the control and treatment groups were not significant on these indicators. This may indicate that despite differences in educational attainment and financial market participation, the two groups showed no significant difference in the proportion of risk asset allocation.

The descriptive analysis reveals that there is a breakpoint in higher education attainment and household financial market participation between the control and treatment groups. This provides intuitive evidence for our study on how changes in higher education influence changes in household financial market participation.

Table 1: Definitions of Variables and Descriptive Statistics

Variables	Definition	Control group	Treatment group	Entire sample
N	Sample size	15159	6881	22040
D_stock	Ownership of stock accounts	0.063 (0.242)	0.136 (0.343)	0.0842 (0.278)
D_all_assets	Participation in financial market investment	0.113 (0.317)	0.210 (0.407)	0.142 (0.349)
Equity_share	Proportion of equity assets to financial assets	0.010 (0.067)	0.023 (0.097)	0.014 (0.078)
All_assets_share	Proportion of financial market risk assets to financial assets	0.159 (0.275)	0.171 (0.269)	0.162 (0.274)
College	1 represents education level of junior college or higher; 0 represents education level below junior college	0.148 (0.355)	0.378 (0.485)	0.172 (0.377)
Score	Financial literacy level	0.546 (0.808)	0.571 (0.920)	0.551 (0.829)
Age	Age	49.150 (5.394)	30.210 (4.282)	43.560 (10.030)
Gender	1 represents male; 0 represents female	0.781 (0.414)	0.696 (0.460)	0.756 (0.430)
House	Whether owning two or more residential properties: 1 indicates yes, otherwise 0	0.278 (0.448)	0.378 (0.485)	0.308 (0.461)
Marriage	1 represents married; 0 represents unmarried, separated, divorced, etc	0.903 (0.297)	0.738 (0.440)	0.854 (0.353)
Rural	1 represents rural family; 0 represents urban family,	0.366 (0.482)	0.121 (0.326)	0.294 (0.456)
Ln(Income)	logarithm of head of household's income	10.770 (1.415)	11.100 (1.434)	10.870 (1.428)
Ln(Asset)	logarithm of family assets.	12.980 (1.508)	11.830 (2.297)	12.640 (1.853)

Note: The data is computed based on CHFS 2015, 2017, and 2019 data.

4.3 Graph analysis

As shown in Figure 2, each point in the graph represents the proportion of the sample from a given birth year that attained higher education, while the lines represent the quadratic polynomial fitting curves between birth year and higher education attainment rates. Around the 1982 cutoff point, there is a clear jump in higher education attainment rates between the control group and the treatment group. Specifically, the treatment group (individuals born in 1982 or later), influenced by the college enrollment expansion policy, shows a significantly higher probability of attaining higher education. This jump indicates that the college enrollment expansion policy likely played a significant role in improving educational attainment levels.

Figure 3 illustrates the relationship between birth year and stock market participation rates among samples born around 1982. Each point represents the stock market participation rate for individuals of a specific birth year, while the lines show the quadratic fitting curves between birth year and stock market participation rates. From Figure 3, a clear jump in stock market participation rates can be observed around the 1982 cutoff point, suggesting that the college enrollment expansion policy may have increased the proportion of individuals participating in the stock market.

As shown in Figure 4, financial market participation rates also exhibit a significant jump near the

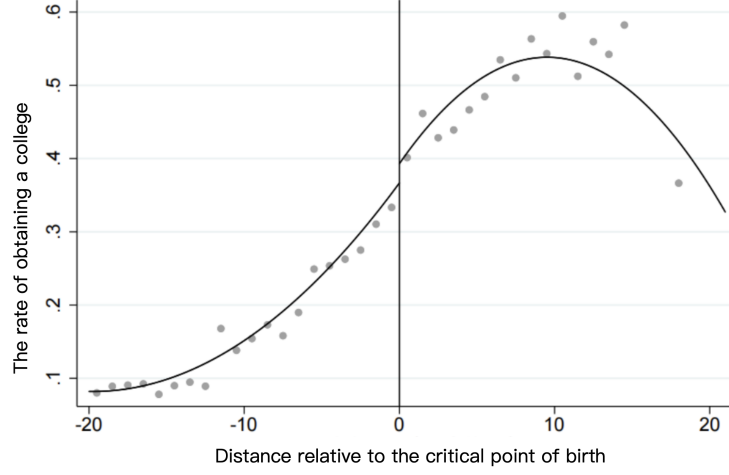


Figure 2: The breakpoint in the higher education attainment rate for individuals born before and after the critical point affected by the expansion of university admissions

policy cutoff point. This indicates that individuals who received higher education are more inclined to participate in financial markets. These figures provide intuitive evidence that the college enrollment expansion policy may have positively influenced individuals' financial market participation behavior.

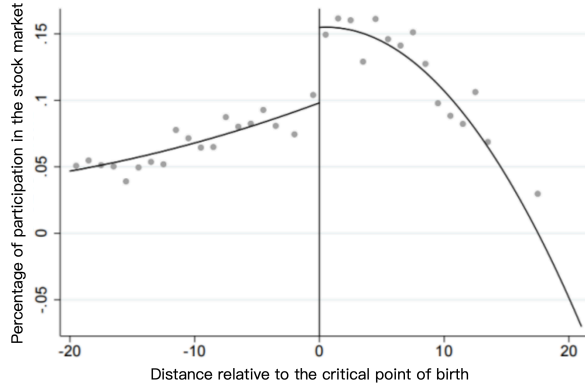


Figure 3: The breakpoints in the stock market participation ratio for samples born before and after the critical point affected by the university expansion policy

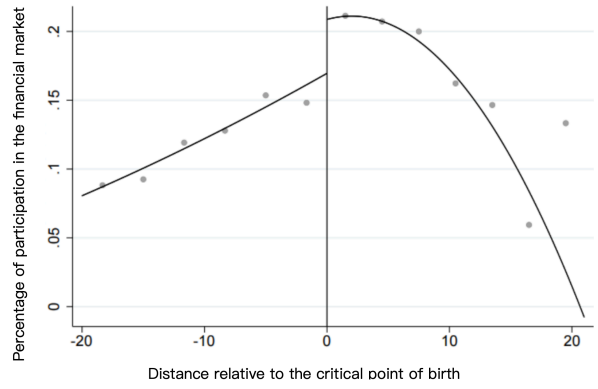


Figure 4: The breakpoints in the financial market participation ratio for samples born before and after the critical point affected by the university expansion policy

As shown in Figure 5, there is no significant change in the proportion of stock assets around the 1982 cutoff point, suggesting that the college enrollment expansion policy may have had a limited impact on stock asset allocation. In contrast, Figure 6 focuses on the proportion of risk assets in total financial assets. Unlike Figure 5, a more noticeable jump in the proportion of risk assets can be observed near the policy cutoff point. This may indicate that individuals with higher education exhibit a shift in preference toward risk assets in their financial asset allocation. These graphical analyses provide an intuitive perspective and preliminary evidence for further investigation into the impact of the college enrollment expansion policy on individual financial asset allocation behavior.

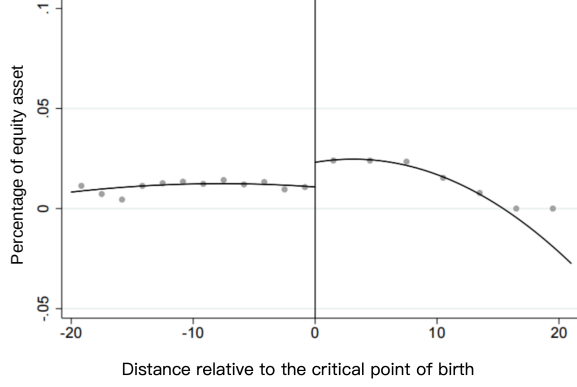


Figure 5: The breakpoints in percentage of equity asset allocation for samples born before and after the critical point affected by the university expansion policy

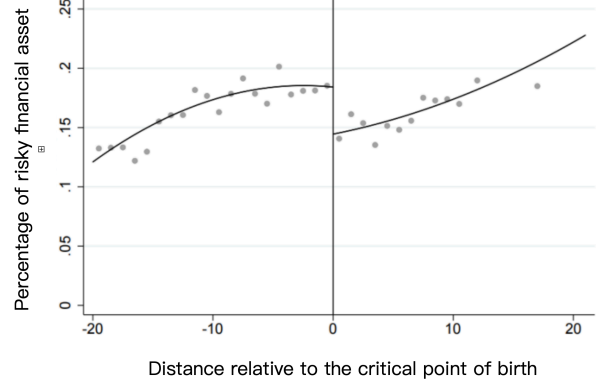


Figure 6: The breakpoints in percentage of risky financial asset allocation for samples born before and after the critical point affected by the university expansion policy

4.4 Empirical results

4.4.1 The Impact of University Enrollment Expansion on Household Financial Market Participation

Table 2 presents the reduced form estimation results for the impact of university enrollment expansion on household financial market participation. The estimation results indicate that, under the optimal window width condition², the implementation of university enrollment expansion policy has a significant effect on household financial market participation rates.

Regarding the estimated impact of university enrollment expansion on the ownership of stock accounts, whether or not control variables are included, the effect of university enrollment expansion on stock account ownership is consistently positive and statistically significant at the 1% significance level. Without including control variables, the implementation of university enrollment expansion policies leads to an average increase of 6% in the probability of households in the treatment group owning stocks compared to the control group. After incorporating baseline control variables, this estimate increases to 7%.

The estimation results for the impact of university enrollment expansion on participation in the financial market suggest a positive effect. Without including baseline control variables, the implementation of university enrollment expansion policies increases the probability of households in the treatment group participating in the formal financial market by an average of 5% compared to the control group. After including baseline control variables, this probability increases to 9%.

Table 2: The Impact of University Enrollment Expansion on Household Financial Market Participation

	The impact of university enrollment expansion on the ownership of stock accounts		The impact of university enrollment expansion on the financial market participation	
expansion	0.0607*** (3.60)	0.0770*** (4.93)	0.0505* (2.40)	0.0934*** (4.71)
observation	8280	7921	7509	7137
optimal window width	YES	YES	YES	YES
controls	NO	YES	NO	YES

Note: Values in parentheses represent standard deviations; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Control variables include gender (male=1), marital status (married=1), rural/urban status (rural=1), logarithm of household head's income, and logarithm of household assets.

Table 3 presents the estimation results of the impact of higher education on financial market participation using the Two-Stage Least Squares (2SLS) method. The first-stage regression results indicate that

²This study employs the default settings in Stata, the Mean Squared Error (MSE) optimal bandwidth method and the rectangular kernel function, to select the optimal bandwidth.

the college enrollment expansion policy significantly increased the likelihood of attending college among the treatment group. After including control variables, the probability of college enrollment becomes to 11.5% and 11.3%, and it holds at a significance level of 1%. This finding aligns with reports from the China Household Finance Survey and Research Center, confirming that the college enrollment expansion policy effectively improved educational levels.

The second-stage regression results show that higher education has a significant positive effect on stock market participation. Compared to individuals without a college education, families with higher education have, on average, a 67.4% higher probability of participating in the stock market. Even after adding control variables, this proportion only slightly decreases to 66.7%, suggesting that other characteristic variables have minimal influence on the results, thus supporting hypothesis **H1**.

It also can be observed that, relative to individuals who have not received university education, households with university education have, on average, a 77.1% higher probability of participating in the stock market. After incorporating control variables, this estimate increases to 82.8%. Hypothesis **H2** is confirmed.

The estimation results for the impact of university education on household financial market participation suggest that, under the optimal window width condition, regardless of whether factors such as age, gender, marital status, and rural/urban status are controlled for, university education significantly encourages households to participate more in the financial market.

Table 3: 2SLS Estimation of the Impact of Higher Education on Financial Market Participation

	The impact of higher education on the ownership of stock accounts		The impact of higher education on participation in financial markets	
	college			
	0.674**	0.667***	0.771*	0.828***
	(2.96)	(3.98)	(2.11)	(3.80)
The impact of university enrollment expansion on obtaining higher education				
expansion	0.090***	0.115***	0.065**	0.113***
	(0.026)	(0.021)	(0.026)	(0.023)
observation	8280	7921	7509	7137
optimal window width	YES	YES	YES	YES
controls	NO	YES	NO	YES

Note: Values in parentheses represent standard deviations; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Control variables include gender (male=1), marital status (married=1), rural/urban status (rural=1), logarithm of household head's income, and logarithm of household assets.

4.4.2 The Impact of University Enrollment Expansion on Household Allocation of Risk Assets

Table 4 presents the simplified equation estimation results for the impact of university enrollment expansion on household risk asset allocation. The estimation results suggest that under the optimal window width condition, the implementation of university enrollment expansion policies has a relatively small effect on household risk asset allocation.

In the estimation results regarding the impact of university enrollment expansion on the proportion of stock assets in financial assets, whether or not control variables are included, the effect of university enrollment expansion on stock account ownership is small and not statistically significant at a lower significance level. Without including baseline control variables, the implementation of university enrollment expansion policies led to an average increase of 0.9% in the proportion of stock assets in financial assets for the treatment group compared to the control group, which is statistically significant at the 10% significance level. After incorporating baseline control variables, this estimate increased to 1% and was statistically significant at the 5% significance level.

However, the estimation results for the impact of university enrollment expansion on the proportion of risk assets within financial assets suggest that university enrollment expansion does not have a significant effect on the allocation of risk assets in the sample population. Without including baseline control variables, the implementation of university enrollment expansion policies led to a decrease in the proportion of risk assets within financial assets for the treatment group compared to the control group. After incorporating baseline control variables, the impact of university enrollment expansion on the proportion of risk assets within financial assets was positive but did not pass the significance test.

Table 4: The Impact of University Enrollment Expansion on Household Allocation of Risk Assets

	The impact of university enrollment expansion on the proportion of stock assets in financial assets		The influence of university enrollment expansion on the proportion of risky assets in financial assets	
expansion	0.00987*	0.0142**	-0.0344*	0.0146
	(2.12)	(3.04)	(-2.02)	(0.86)
Observation	6382	6067	7493	7131
Optimal window width	YES	YES	YES	YES
Controls	NO	YES	NO	YES

Note: Values in parentheses represent standard deviations; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Control variables include gender (male=1), marital status (married=1), rural/urban status (rural=1), logarithm of household head's income, and logarithm of household assets.

Table 5 reports the 2SLS estimation results for the impact of university education on the proportion of household stock assets. The first-stage estimation results indicate that university enrollment expansion has a significant impact on the likelihood of obtaining a university education. However, the 2SLS estimation results suggest that university education does not have a significant impact on the average proportion of household financial assets invested in stocks. This suggests that while households with university education are more likely to participate in the stock market, the proportion of their financial assets invested in stocks does not significantly increase after their participation in the stock market. Hypothesis **H3** is not supported.

Besides, regardless of whether baseline control variables are included, higher education does not have a significant impact on the average proportion of household financial assets invested in risk assets. Hypothesis **H4** is not supported either. The estimation results regarding the impact of university education on household risk asset allocation indicate that, under the optimal window width condition, there is no significant evidence to suggest that university education has a significant impact on the proportion of stock assets or risk assets in the household portfolio.

These conclusions suggest that while university education increases financial market participation, it does not significantly alter the allocation of stock assets or risk assets within the household's financial portfolio.

Table 5: 2SLS Estimation of the Impact of Higher Education on the Proportion of Household Stock Asset Allocation

	The impact of higher education on the proportion of stock assets to financial assets		The impact of higher education on the proportion of risk assets to financial assets	
college	0.179	0.147*	-0.557	0.137
	(1.45)	(2.41)	(-1.36)	(0.86)
	0.147	0.016	0.175	0.392
	The impact of university expansion on obtaining higher education			
expansion	0.055*	0.096***	0.062**	0.106***
	(0.033)	(0.028)	(0.029)	(0.025)
observation	6382	6067	7493	7131
optimal window width	YES	YES	YES	YES
Controls	NO	YES	NO	YES

Note: Values in parentheses represent standard deviations; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Control variables include gender (male=1), marital status (married=1), rural/urban status (rural=1), logarithm of household head's income, and logarithm of household assets.

5 Validity Testing

5.1 Randomness of the Treatment Variable

In this study, the cutoff point is set at individuals born in 1982. The validity of this cutoff in satisfying the assumption of independence in assignment rests on several factors. First, the running variable employed in this paper is the individual’s year of birth, which is a naturally occurring attribute and not subject to personal choice. Neither individuals nor their parents can manipulate or predict the timing of birth to align with the 1999 higher education expansion policy. Second, the policy was proposed and implemented within a relatively short time frame, leaving limited room for anticipation or behavioral adjustment by households. This reduces the likelihood that families could plan childbirth based on expectations of the upcoming reform. Third, China’s household registration system restricts arbitrary changes to recorded birth dates, further reinforcing the credibility of birth year as a valid assignment variable. Fourth, the higher education expansion policy was a centrally determined national initiative, independent of individual choices. This ensures the exogeneity of the policy’s implementation and supports the assumption that individuals near the cutoff were exposed to the policy in a quasi-random manner.

Taken together, these factors suggest that the treatment variable in this study exhibits strong randomness, providing a solid foundation for identifying the causal effect of the 1999 higher education expansion through a regression discontinuity design. The exogeneity requirement of the RD design ensures that individuals close to the cutoff are as-if randomly assigned, allowing for credible estimation of the true impact of education on financial market participation.

5.2 Continuity test of control variables

To ensure that the observed discontinuity in the outcome variable is attributable to the key explanatory variable—namely, educational attainment—rather than to other covariates, it is essential that all control variables remain continuous around the cutoff. In other words, except for the college enrollment rate, which is expected to exhibit a discontinuity at the threshold, other control variables such as age, gender, and marital status should not display significant changes around the cutoff point.

Table 6 presents the results of the continuity test under the optimal bandwidth. The findings indicate that the continuity assumption cannot be rejected for the baseline control variables at the cutoff, suggesting that their distributions do not differ significantly on either side of the threshold. This supports the validity of the regression discontinuity design. Although asset holdings show significance at the 10% level, this does not necessarily imply a substantial influence on the outcome variable. Further robustness checks are required to assess the stability of the results.

Table 6: Continuity Tests for Control Variables

Dependent variable	Gender	Marriage	Rural	House	Income	Financial assets
expansion	-0.351 (-0.83)	0.0748 (0.23)	-0.908 (-1.42)	0.324 (0.54)	0.360 (0.23)	-25.96* (-2.15)
observations	7539	5293	5293	5293	6153	7411

Note: Values in parentheses represent standard deviations; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

5.3 Placebo test

To ensure that the observed increase in household participation in financial markets is indeed driven by the implementation of the higher education expansion policy, this study conducts a placebo test. Specifically,

we select two non-policy years—1995 and 2002—as hypothetical policy implementation points to examine whether similar policy effects emerge at these alternative cutoffs. The purpose of this test is to verify whether the observed effects around the actual cutoff are unique to the 1999 expansion of higher education.

If no significant discontinuity in college enrollment rates is observed at these placebo cutoff points, it would provide further evidence that the estimated effects are genuinely attributable to the 1999 policy intervention. Table 7 presents the estimation results of the continuity test in college enrollment using the two placebo years as hypothetical thresholds.

The results presented in Table 7 indicate that, after incorporating baseline control variables and applying different bandwidth selections, no significant discontinuity in college attainment is observed when the policy implementation is hypothetically assumed to have occurred earlier or later³.

This finding further supports the main hypothesis: the observed discontinuity in college attainment among individuals born around 1982 is indeed driven by the 1999 higher education expansion policy. The placebo test thus reinforces the reliability and validity of the study’s conclusions.

Table 7: Testing the Continuity of University Education Attainment Rates Using Assumed Years of University Expansion

University expansion	The assumed cutoff year of 1995 as the breakpoint		The assumed cutoff year of 2002 as the breakpoint	
Optimal bandwidth for processing effect	0.00302 (0.06)	0.00272 (0.06)	0.0438 (0.96)	0.0438 (0.96)
0.5 times the optimal processing effect	0.00902 (0.32)	0.00902 (0.32)	0.0106 (0.40)	0.0106 (0.40)
2 times the optimal processing effect	-0.0316 (-0.93)	-0.00735 (-0.16)	-0.0205 (-0.64)	-0.0205 (-0.64)
Observations	3369	5737	4187	4187
Controls	NO	YES	NO	YES

Note: Values in parentheses represent standard deviations; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Baseline control variables include gender (male=1), marital status (married=1), rural/urban status (rural=1), logarithm of household head’s income, and logarithm of household assets.

5.4 Testing for Sample Selection Dependence

The sample selection dependence test, commonly referred to as the “donut-hole test,” is a robustness check used to assess the reliability of regression discontinuity results. The core idea is to exclude observations close to the cutoff point in order to test whether these observations disproportionately influence the regression outcomes—particularly when there is concern that such samples may be subject to manipulation. In this study, we exclude individuals born between 1980 and 1983 from the analysis as part of this robustness check.

Columns (1) and (2) of Table 8 report the estimated effects of the higher education expansion on household participation in financial markets after removing observations near the cutoff. The results remain positive and statistically significant at the 5% level, and the estimates are largely consistent with those obtained when these observations are included. This suggests that the higher education expansion significantly increased household participation in stock and financial markets, even when accounting for potential sample manipulation.

Columns (3) and (4) of Table 8 present the estimated effects of the expansion on household risk asset allocation, again after excluding observations near the cutoff. The results show no significant effect of college education on the share of stock assets or overall risky asset allocation, which is consistent with the estimates using the full sample. These findings further support the robustness of the main results.

³To ensure sufficient sample size and meet the requirements of the placebo test, we use data from three years before and after each hypothetical policy change point.

Table 8: The Impact of University Enrollment Expansion on Household Financial Assets Allocation

	(1) The impact of university enrollment expansion on the ownership of stock accounts		(2) The impact of university enrollment expansion on participation in financial markets		(3) The impact of university enrollment expansion on the proportion of stock assets in financial assets		(4) The influence of university enrollment expansion on the proportion of risky assets in financial assets	
expansion	0.0954** (3.12)	0.120*** (4.38)	0.0711 (1.72)	0.133*** (3.37)	0.0192 (1.34)	0.0384** (2.98)	-0.0114 (-0.29)	0.0478 (1.24)
observation	6055	5787	5173	4911	4051	3843	5165	4907
optimal window width	YES	YES	YES	YES	YES	YES	YES	YES
controls	NO	YES	NO	YES	NO	YES	NO	YES

Note: Values in parentheses represent standard deviations; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Baseline control variables include gender (male=1), marital status (married=1), rural/urban status (rural=1), logarithm of household head's income, and logarithm of household assets.

6 Mechanism Analysis

6.1 Financial cognitive level

In measuring financial literacy, this study adopts the method proposed by [Yin et al. \(2019\)](#), using three key survey questions—on interest rate calculation, understanding of inflation, and understanding of investment risk—to construct a financial literacy index. Specifically, the number of correct answers to these three questions serves as the measure of financial literacy, with one point assigned for each correct response.

As shown in Table 9, after incorporating financial literacy into the model, the coefficient on financial literacy is consistently around 0.066 and statistically significant at the 1% level. This indicates that the higher education expansion policy may influence household stock ownership and financial market participation through the channel of enhanced financial literacy, thus providing empirical support for Hypotheses H1a and H2a. These results not only reaffirm the positive impact of the higher education expansion on household financial market participation, but also highlight the critical role of improving financial literacy in promoting such engagement.

Table 9: Impact of University Enrollment Expansion on Household Financial Market Participation

	The Impact of university enrollment expansion on stock account ownership	The impact of university enrollment expansion on financial market participation
expansion	0.0395*** (9.70)	0.0392*** (9.60)
financial cognitive level	0.0655*** (34.19)	0.0656*** (34.14)
Observations	12835	12859
Controls	YES	YES

Note: Values in parentheses represent standard deviations; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Baseline control variables include gender (male=1), marital status (married=1), rural/urban status (rural=1), and logarithm of household assets.

6.2 Income level

Table 10 reports the reduced-form results after controlling for household income. When accounting for income levels, the positive effect of higher education expansion on financial market participation remains statistically significant. This finding aligns with the results of [Liu and Hu \(2018\)](#), who suggest that households with higher income levels tend to exhibit a stronger willingness to participate in financial markets.

The regression results show that, after controlling for other relevant covariates, the implementation of the higher education expansion policy led to an average increase of 7% in the probability of owning a stock account and a 9% increase in the likelihood of participating in financial markets for the treatment group relative to the control group. Additionally, household income is statistically significant at the 1%

level. These findings underscore the critical influence of income level on financial market participation, thereby providing empirical support for Hypotheses H1b and H2b.

In summary, the expansion of higher education may promote more active household participation in financial markets by improving income levels.

Table 10: The Impact of University Enrollment Expansion on Household Financial Market Participation

	The impact of university enrollment expansion on stock account ownership	The impact of university enrollment expansion on financial market participation
expansion	0.0712*** (15.68)	0.0994*** (18.19)
Ln(Income level)	0.0227*** (15.82)	0.0337*** (19.46)
Observations	20815	21140
Other controls	YES	YES

Note: Values in parentheses represent standard deviations; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Baseline control variables include gender (male=1), marital status (married=1), rural/urban status (rural=1), and logarithm of household assets.

7 Heterogeneity Analysis

7.1 Urban-Rural Heterogeneity

The persistence of China’s long-standing dual economic structure has resulted in substantial disparities in household allocation of risky financial assets between urban and rural areas. As shown in Table 11, urban households exhibit significantly higher participation rates in both the stock market and broader financial markets compared to their rural counterparts. This highlights the pronounced urban-rural divide in household financial market participation in China. Therefore, it is essential to account for urban-rural heterogeneity when examining the impact of higher education on household financial market participation.

First, urban areas generally have better access to high-quality educational resources, including more developed higher education institutions, better-qualified teachers, and greater educational investment. As a result, urban residents are more likely to benefit from the expansion of higher education, which may, in turn, influence their financial market behavior. Second, differences in the economic foundations between urban and rural areas lead to significant variation in investable household assets. While higher education expansion may enhance individual and household income levels and thereby promote financial market participation, this effect is likely to be more pronounced among urban households with stronger economic bases. Finally, urban households typically have greater access to a wider range of financial products and services and tend to possess higher levels of financial knowledge and awareness, which facilitates more active participation in financial markets. In contrast, financial literacy and service coverage in rural areas tend to lag behind, potentially leading to heterogeneous regression outcomes across urban and rural groups.

Table 11: Household Financial Asset Allocation by Urban and Rural Areas

Asset Categories	Urban	Rural	Total participation rate %
	Participation rate %	Participation rate %	
Stock	12.0	0.4	8.6
Financial markets	19.2	2.3	14.2
N	15223	6415	21638

Note: Data source: Computed based on CHFS (China Household Finance Survey) data from the years 2015, 2017, and 2019.

The results presented in Table 12 clearly reveal that the impact of the higher education expansion policy differs between urban and rural households. Among urban households, the policy has a significantly positive effect on stock ownership. In contrast, this effect is not statistically significant for rural households and even shows a negative trend. When comparing the coefficient estimates for urban households with those for the full sample, no significant difference is observed, suggesting that the positive impact of the higher education expansion on household stock ownership is primarily driven by urban households.

Similarly, in terms of financial market participation, urban households experience a significant positive effect from the higher education expansion, whereas rural households show no statistically significant change. The similarity between the coefficients for urban households and those in the full sample further underscores that the policy’s effect in promoting risky financial asset allocation is concentrated among urban households.

These findings highlight the persistent inequalities in education and financial inclusion between urban and rural areas in China. The expansion of higher education appears to have promoted more active financial market participation among urban households by improving financial literacy and enhancing household economic conditions. In contrast, rural areas have not benefited to the same extent from the policy, potentially due to the unequal distribution of educational resources, lower levels of financial knowledge, or limited economic capacity. These results suggest that policymakers should adopt more targeted measures to promote inclusive development in both education and finance, thereby narrowing the urban-rural divide.

Table 12: Impact of University Enrollment Expansion on Household Financial Market Participation by Urban and Rural Area

	The impact of university enrollment expansion on the ownership of stock accounts in urban and rural areas				The impact of university enrollment expansion on the financial market participation in urban and rural areas			
	Urban		Rural		Urban		Rural	
expansion	0.0689*** (3.63)	0.0979*** (5.19)	-0.0233 (-1.18)	-0.0142 (-0.81)	0.0579* (2.47)	0.118*** (4.84)	-0.00769 (-0.31)	-0.003 (-0.81)
observation	7514	7212	947	1106	7721	6569	1054	764
optimal window width	YES	YES	YES	YES	YES	YES	YES	YES
controls	NO	YES	NO	YES	NO	YES	NO	YES

Note: Values in parentheses represent standard deviations; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Control variables include gender (male=1), marital status (married=1), rural/urban status (rural=1), logarithm of household head’s income, and logarithm of household assets.

To further interpret the regression results, the significant positive response of urban households to the higher education expansion policy—contrasted with the absence of a comparable effect among rural households—can be explained by several factors:

Income disparities affect investment willingness: As discussed in the mechanism analysis, households with higher income levels are more likely to invest in risky financial assets. According to the China Household Finance Survey, the average annual income of urban households is significantly higher than that of rural households. This income gap may be one of the key reasons why the higher education expansion policy significantly increased financial market participation among urban households but had a limited effect on rural households.

Differences in access to information: [Li et al. \(2020\)](#) suggest that households’ ability to access private financial market information or fully utilize publicly available information significantly enhances their likelihood of participating in financial markets. Due to factors such as geographic location and interpersonal networks, urban households are generally better positioned to acquire both private and public financial information, which promotes more active engagement in financial markets.

Social networks and risk preferences: [Wang et al. \(2015\)](#) find that social networks play a buffering role in households’ investment decisions by helping to mitigate financial risks, thereby influencing individual risk preferences. The differences in social support networks between urban and rural households—particularly regarding economic assistance—may mean that urban households receive more support in the

face of potential investment losses, making them more willing to take on financial risks and participate in the stock market. In summary, the significant disparities between urban and rural areas in income levels, access to financial information, social network support, and overall wealth jointly contribute to the observed differences in household financial market participation. These factors help explain why the higher education expansion policy had a pronounced positive effect on urban households but little impact on rural ones. These findings underscore the importance of accounting for urban-rural heterogeneity in policy design to ensure more equitable and effective outcomes.

7.2 Heterogeneity Analysis by Property Ownership

The regression results presented in Table 13 illustrate the heterogeneity in the impact of the higher education expansion policy on household financial market participation based on the number of properties owned. Among households owning a single property, the policy significantly increased participation in the stock market. However, for households owning two or more properties, the effect of the policy is not statistically significant.

These findings suggest that the effect of the higher education expansion on promoting household participation in financial markets varies notably across different levels of property ownership.

Table 13: Impact of College Expansion on Household Stock Ownership by Number of Housing Properties Owned

	The impact of university enrollment expansion on the ownership of stock accounts				The impact of university enrollment expansion on the financial market participation			
	One house		Two or more house		One house		Two or more house	
expansion	0.0758*** (4.07)	0.0927*** (5.55)	-0.0132 (-0.26)	0.0264 (0.55)	0.0485* (2.25)	0.0843*** (3.84)	0.0582 (1.38)	0.139** (3.03)
observations	4874	5326	1582	1523	5819	5532	2789	1928
optimal window width	YES	YES	YES	YES	YES	YES	YES	YES
controls	NO	YES	NO	YES	NO	YES	NO	YES

Note: Values in parentheses represent standard deviations; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Control variables include gender (male=1), marital status (married=1), rural/urban status (rural=1), logarithm of household head's income, and logarithm of household assets.

The observed heterogeneity in the effect of the higher education expansion policy on household financial market participation across households with different levels of property ownership may be attributed to the perception of real estate and stocks as substitute investment options. The continuous rise in housing prices in China has objectively contributed to the formation of a “safe asset illusion” regarding real estate among Chinese residents, which, in turn, influences household investment behavior. As shown in Table 14, the proportion of households owning two or more properties has been increasing, suggesting that an increasing number of families may regard real estate as a “safe” investment asset, leading to asset concentration in property and a substitution away from riskier financial investments.

Second, real estate investment typically requires a large capital outlay and is relatively illiquid. For households owning only one property, the high costs and low liquidity associated with additional real estate investment may deter further investment in property. Instead, these households may allocate available funds to the stock market, which offers greater liquidity and potentially higher returns despite greater risk. When the return on property investment appears higher or more stable, households may be inclined to increase investment in real estate while reducing allocation to risky assets such as stocks (as shown in Table 15).

8 Conclusion

Drawing on data from the China Household Finance Survey (CHFS) for the years 2015, 2017, and 2019, this paper employs a fuzzy RDD using the 1999 higher education expansion policy as an exogenous shock

Table 14: Percentage of Households with Different Numbers of Housing Properties Owned

	Control group	Experimental group
One	72.4	63.0
Two or more	27.6	37.0
Observations	15159	6881

Source: "China Household Finance Survey Report".

Table 15: Acquisition Cost and Current Value of Housing (in thousands of yuan)

	Urban			Rural		
	The first property	The second property	The third property	The first property	The second property	The third property
Cost price	19.10	39.33	62.03	6.28	16.39	22.75
Current value	84.10	95.67	122.01	18.34	31.68	40.34

Source: China Household Finance Survey Report.

to examine the causal effect of higher education on household risky financial asset investment. Specifically, household financial market participation and the share of risky asset allocation are used as key outcome variables. Both reduced-form estimates and two-stage least squares (2SLS) results reveal that higher education significantly increases the likelihood of household participation in financial markets. However, it does not significantly affect the proportion of risky financial assets held by households. This indicates that while higher education can broaden the base of market participants, it does not necessarily deepen household risk-taking in terms of asset allocation.

Mechanism analysis shows that higher education promotes financial market participation primarily through improved financial literacy and increased household income. This suggests two policy implications. First, in addition to expanding higher education access, the government should invest in nationwide financial education programs. Financial literacy training should be integrated into both formal education curricula and adult education initiatives to equip citizens with the skills to understand and navigate financial products and markets. Second, policies that enhance household income—especially for disadvantaged groups—can indirectly foster financial inclusion by increasing households' investment capacity and willingness to engage with financial markets.

The study's heterogeneity analysis further reveals that the effect of higher education on financial market participation is not uniform across all households. Urban households benefit significantly from the education reform, while rural households show no statistically significant response. This urban-rural divide may be driven by disparities in income levels, access to financial information and services, and differences in social network support and risk tolerance. Similarly, the effect of higher education varies by real estate ownership. The impact is stronger among households owning one property but becomes insignificant among those owning two or more. This may reflect the perceived substitutability between real estate and risky financial assets, as well as varying risk preferences and liquidity constraints. As housing becomes an increasingly dominant investment vehicle—perceived by many as a safe asset—households with multiple properties may opt out of the financial market, even if they have attained higher education.

These findings underscore the need for targeted policy design. Policymakers should address the structural barriers faced by rural households, including limited access to financial services, insufficient financial literacy, and constrained income mobility. Efforts should be made to expand digital financial infrastructure in rural regions, improve access to diversified investment instruments, and ensure rural populations benefit equitably from educational reforms. Additionally, tax incentives or financial instruments tailored to encourage moderate engagement with risky asset markets among high-asset households could help rebalance household investment portfolios.

References

- Ashenfelter, O. and Rouse, C. (1998). Income, schooling, and ability: Evidence from a new sample of identical twins. *The Quarterly Journal of Economics*, 113(1):253–284.
- Badarinza, C., Campbell, J. Y., and Ramadorai, T. (2016). International comparative household finance. *Annual Review of Economics*, 8:111–144.
- Black, S. E., Devereux, P. J., Lundborg, P., and Majlesi, K. (2018). Learning to take risks? the effect of education on risk-taking in financial markets. *Review of Finance*, 22(3):951–975.
- Bodie, Z., Merton, R. C., and Samuelson, W. F. (1992). Labor supply flexibility and portfolio choice in a life cycle model. *Journal of Economic Dynamics and Control*, 16(3-4):427–449.
- Briggs, J., Cesarini, D., Lindqvist, E., and Östling, R. (2021). Windfall gains and stock market participation. *Journal of Financial Economics*, 139(1):57–83.
- Calvet, L. E. and Sodini, P. (2014). Twin picks: Disentangling the determinants of risk-taking in household portfolios. *The Journal of Finance*, 69(2):867–906.
- Campbell, J. Y. (2006). Household finance. *The Journal of Finance*, 61(4):1553–1604.
- Christiansen, C., Joensen, J. S., and Rangvid, J. (2008). Are economists more likely to hold stocks? *Review of Finance*, 12(3):465–496.
- Cole, S., Paulson, A., and Shastry, G. K. (2014). Smart money? the effect of education on financial outcomes. *The Review of Financial Studies*, 27(7):2022–2051.
- Cooper, R. and Zhu, G. (2016). Household finance over the life-cycle: What does education contribute? *Review of Economic Dynamics*, 20:63–89.
- Deng, X., Cheng, X., and Fu, Z. (2021). Household financial decision-making and macroeconomic fluctuations. *Emerging Markets Finance and Trade*, 57(14):4143–4165.
- Dohmen, T., Falk, A., Huffman, D., and Sunde, U. (2010). Are risk aversion and ompatience related to cognitive ability? *American Economic Review*, 100(3):1238–1260.
- Frederick, S. (2005). Cognitive reflection and decision making. *Journal of Economic Perspectives*, 19(4):25–42.
- Gan, L., Yin, Z., and Tan, J. (2015). China household finance survey report 2014. *Southwestern University of Finance and Economics Press: Chengdu, China*.
- Gray, D., Montagnoli, A., and Moro, M. (2021). Does education improve financial behaviors? quasi-experimental evidence from britain. *Journal of Economic Behavior & Organization*, 183:481–507.
- Hermansson, C. and Jonsson, S. (2021). The impact of financial literacy and financial interest on risk tolerance. *Journal of Behavioral and Experimental Finance*, 29:100450.
- Li, J., Li, X., and Qi, L. (2020). Private information public information and financial market participation in china. *Journal of Financial Research (China)*, 478(4):147–165.
- Liu, S. and Hu, A. (2018). Returns to college education: A natural experiment from expansion of higher education in china. *Studies in Labor Economics(China)*, 4:48–70.

- Lusardi, A., Michaud, P.-C., and Mitchell, O. S. (2017). Optimal financial knowledge and wealth inequality. *Journal of Political Economy*, 125(2):431–477.
- Qiao, Y. and Cai, Y. (2023). Financial assets and happiness: evidence from the china household finance survey. *Applied Economics Letters*, 30(4):466–471.
- Ren, C., X, X., and Zhou, Y. (2020). Can education affect household stock market participation behaviors? evidence from china’s compulsory schooling law. *China Journal of Economics*, 7(4):156–188.
- Tsang, A. and Yu, L. (2023). Socially responsible local firms and stock market participation: Evidence from the u.s. household survey. *Journal of Behavioral and Experimental Finance*, 40:100854.
- Van Rooij, M., Lusardi, A., and Alessie, R. (2011). Financial literacy and stock market participation. *Journal of Financial economics*, 101(2):449–472.
- Vissing-Jorgensen, A. (2002). Towards an explanation of household portfolio choice heterogeneity: Non-financial income and participation cost structures.
- Wang, C., Chai, S., Tian, C., and Wu, S. (2015). Family social network and stock market participation. *J World Econ(China)*, 38:105–24.
- Wang, C. and Tian, C. (2012). Stock market participation, participation rate and determinants. *Economic Research Journal(China)*, 10:97–107.
- Wu, W. and Shen, T. (2015). Cohort effect of education and stock market participation. *Journal of Financial Research(China)*, (8):175–190.
- Xing, L., Han, D., and Xie, H. (2023). Government digital policy breaks the mystery of “limited participation” in china’s home finance market. *Scientific Reports*, 13(1):20233.
- Yin, Z., Song, Q., and Wu, Y. (2014). Financial literacy, trading experience and household portfolio choice. *Economic Research Journal(China)*, 49(04):62–75.
- Yin, Z., Yue, P., and Chen, X. (2019). Financial market participation risk heterogeneity and household well being. *Journal of Financial Research(China)*, 466(4):168–187.
- Zong, Q., Liu, C., and Zhou, Y. (2015). Social endowment insurance and risky financial assets investment of households in china: Evidence from chfs. *Journal of Financial Research(China)*, 10(424):99–114.